



Parallel multi-fidelity expected improvement method for efficient global optimization

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Abstract

Multi-fidelity optimization (MFO) has received extensive attentions in engineering design, which resorts to augmenting the small number of expensive high-fidelity (HF) samples by a large number of low-fidelity (LF) but cheap samples to improve the optimization performance. A key factor that influences the effectiveness of MFO is how to adaptively assign samples for HF and LF simulations in the iteration process. To address such sample assignment issue in MFO, we propose a new infill criterion named Filter-GEI, which imposes an adaptive filter function on top of the generalized expected improvement (GEI) acquisition function. In particular, by taking the correlations between HF and LF models into account, the Filter-GEI can efficiently allocate HF and LF samples to achieve a good balance in between the local and global search. Furthermore, considering parallel computing, the Filter-GEI infills multiple HF and LF samples in each iteration, which can further improve its efficiency as computing power increases. Through tests on five mathematical toy problems and one engineering problem for the turbine blade design, the effectiveness of the proposed algorithm has been well demonstrated.

Keywords Multi-fidelity optimization · Co-kriging surrogate · Generalized expected improvement · Efficient global optimization · Infill-sampling criterion

1 Introduction

The efficient global optimization (EGO) is well known to be sample efficient (Jones and Schonlau 1998), which has achieved tremendous success in solving engineering problems (Song et al. 2016). Basically, an EGO algorithm consists of four steps as (1) initial sampling; (2) surrogate modeling; (3) selection of new sample candidates by using acquisition functions such as the expected improvement (EI); and (4) querying the new samples and updating the surrogate model. Repeat the above processes in (2) to (4) until the termination condition is met.

The surrogate modeling and selection of new samples through acquisition function constitute the two key elements of EGO. Instead of purely using limited available high-fidelity (HF) samples to build a surrogate, a wide attention has been drawn to multi-fidelity surrogates (MFS), which augment the expensive HF samples by a large number of low-fidelity (LF) yet cheap samples to achieve better modeling accuracy. Generally, the MFS models can be categorized into three groups as (1) correlation function based modeling (Lewis 1996; Han et al. 2013), (2) space mapping (Bakr et al. 2001; Leifsson et al.), and (3) co-kriging methods (Kennedy and O'Hagan 2000; Forrester et al. 2007; Qian and Jeff Wu 2008; Park et al. 2018; Rumpfkeil and Beran 2018). In this paper, we focus on using co-kriging to build a new multi-fidelity surrogate-based optimization algorithm.

On the other hand, when the selection of new sample candidates is concerned, we can use different acquisition functions such as EI and the probability of improvement (PI) (Pickett and Turner 2011; Jun 2012). However, different from EGO algorithms that build on single-fidelity surrogate, we have to decide whether to use high- or low-fidelity

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simulations to query the new samples in MFS-based optimization (abbreviated as MFO). Some pioneering works propose to only add the HF samples in the iteration process (Han et al. 2010; Zhou 2016; Bailly and Bailly 2019; Kim et al. 2018; Jo et al. 2016; Courrier et al. 2016). By taking the cost ratio and the similarities between HF and LF models into account, some other works propose to adaptively infill both HF and LF samples in each optimization cycle (Huang et al. 2006; Gratiet and Garnier 2014; Zhang et al. 2018; Hao et al. 2020).

Note that most previous works as noted above only add in one sample per optimization cycle. However, as pointed out by Ginsbourger et al. (2007), the parallelism is crucial to take advantage of the modern distributed systems to accelerate the optimization process. Specifically, when the simulations can be distributed over different processors, the wall-clock time (i.e., the time of total cycles rather than the number of total function calls) can be greatly reduced (Viana et al. 2013). Meanwhile, to reduce the risks brought by unexpected simulation failure, querying multiple samples per optimization cycle is also encouraged in the engineering process. Thus far, there are extensive studies on parallel optimization for single-fidelity surrogate-based optimization (Haftka et al. 2016; Zhan et al. 2017). However, not much has been reported about paralleled optimization in MFO, where the key issue would be how to intelligently assign the high- and low-fidelity sample batches in each optimization cycle.

To address the above sample assignment task in MFO, we propose a new infill criterion named as Filter-GEI. On one hand, in order to balance in between the local exploitation and global exploration (Schonlau 1997), the GEI acquisition function will suggest new infill sample candidates that either close to the optima of the fitted surrogate or in the area of maximum prediction uncertainty. On the other hand, the LF model can more or less capture the global trend of the HF model over space when we decide to use an MFS to solve an optimization problem (Giselle Fernández-Godino et al. 2019). Therefore, we propose an adaptive filter function on top of the GEI, and we call the HF simulation only when the sample candidates have relatively better estimated objective value, while we query the rest of the samples of GEI through LF simulation to explore the searching space globally. With the above, we can achieve a good balance in between local exploitation and global exploration by making full use of the multi-fidelity and parallel computing resources.

The remainder of this paper is organized as follows: Section 2 presents a brief introduction of co-kriging and GEI acquisition function. Then, Section 3 details our proposed parallel infill criterion, Filter-GEI. After that, Section 4 shows the experimental studies on benchmark functions and an engineering problem. Finally, we draw conclusions in Section 5.

2 Preliminaries

As the surrogate model and acquisition function are the two key elements of surrogate-based optimization, we introduce the basics of co-kriging and GEI acquisition function in this section.

2.1 Co-kriging

Co-kriging (Forrester et al. 2007) is a popularly used multi-fidelity surrogate technique, which models both the LF and HF functions as Gaussian processes, of which mathematical expressions can be written as:

$$\begin{aligned} Y_{LF}(\mathbf{x}) &= \mu_{LF} + Z_{LF}(\mathbf{x}) \\ Y_{HF}(\mathbf{x}) &= \mu_{HF} + Z_{HF}(\mathbf{x}) \end{aligned} \quad (1)$$

where μ_{LF} and μ_{HF} denote the mean function values of the LF and HF model; $Z_{LF}(\mathbf{x})$ and $Z_{HF}(\mathbf{x})$ denote the random Gaussian variables. Then, the relations between the HF and LF functions are formulated as:

$$y_{HF}(\mathbf{x}) = \rho \cdot y_{LF}(\mathbf{x}) + Z_{DF}(\mathbf{x}) \quad (2)$$

where ρ is the scaling factor, and $Z_{DF}(\mathbf{x})$ is the discrepancy function (DF) that models the differences between the HF function and the corrected LF function.

For standard Gaussian process, the correlations between samples $\mathbf{x}_1$ and $\mathbf{x}_2$ can be described as:

$$\text{Cov}[Z(\mathbf{x}_1), Z(\mathbf{x}_2)] = \sigma^2 \exp\left(-\sum_{h=1}^d \theta_h \|x_{1,h} - x_{2,h}\|^2\right) \quad (3)$$

where d denotes the dimension of the problem, σ^2 is the process variance, and θ_h represents the hyper-parameter which determines the bumpiness of fitted function curves. For co-kriging, it deals with Gaussian process of multiple fidelity models, the corresponding covariance matrix and correlation vector can be expressed as:

$$C = \begin{pmatrix} \sigma_{LF}^2 R_{LF}(\mathbf{X}_{LF}, \mathbf{X}_{LF}) & \rho \sigma_{LF}^2 R_{LF}(\mathbf{X}_{LF}, \mathbf{X}_{HF}) \\ \rho \sigma_{LF}^2 R_{LF}(\mathbf{X}_{HF}, \mathbf{X}_{LF}) & \rho^2 \sigma_{LF}^2 R_{LF}(\mathbf{X}_{HF}, \mathbf{X}_{HF}) \\ & + \sigma_{DF}^2 R_{DF}(\mathbf{X}_{LF}, \mathbf{X}_{LF}) \end{pmatrix} \quad (4)$$

$$c = \begin{pmatrix} \rho \sigma_{LF}^2 R_{LF}(\mathbf{X}_{LF}, \mathbf{x}^*) \\ \rho^2 \sigma_{LF}^2 R_{LF}(\mathbf{X}_{HF}, \mathbf{x}^*) + \sigma_{DF}^2 R_{DF}(\mathbf{X}_{LF}, \mathbf{x}^*) \end{pmatrix} \quad (5)$$

where θ_{LF} and θ_{DF} are the bumpiness hyperparameters of LF and DF models, respectively. The function prediction and the related prediction uncertainty of co-kriging model at a new point $\mathbf{x}^*$ are:

$$\begin{aligned} \hat{y}_{HF}(\mathbf{x}^*) &= \beta_0 + \mathbf{c}^T \mathbf{C}^{-1} (\mathbf{y}_s - \beta_0 \mathbf{F}) \\ \hat{s}_{HF}^2(\mathbf{x}^*) &= \sigma^2 \{1 - \mathbf{c}^T \mathbf{C}^{-1} \mathbf{c} + (\mathbf{1} - \mathbf{F}^T \mathbf{C}^{-1} \mathbf{c})^2 / \mathbf{F}^T \mathbf{C}^{-1} \mathbf{F}\} \end{aligned} \quad (6)$$

where $\mathbf{F} = \mathbf{1}$ when the regression order is set as 0.

2.2 Generalized expected improvement (GEI)

Assuming $y(\mathbf{x})$ comes from a Gaussian process with the posterior estimate as $y(\mathbf{x}) \sim N(\hat{y}(\mathbf{x}), s^2(\mathbf{x}))$, the basic idea of the EI acquisition function is to measure how much improvement of the objective function can be made with respect to the current best solution $f_{\min}$, which can be expressed as:

$$EI(\mathbf{x}) = \begin{cases} (f_{\min} - \hat{y}(\mathbf{x}))\Phi(u(\mathbf{x})) + s(\mathbf{x})\phi(u(\mathbf{x})) & (s > 0) \\ 0 & (s \leq 0) \end{cases}$$

$$u(\mathbf{x}) = (f_{\min} - \hat{y}(\mathbf{x}))/s(\mathbf{x}) \quad (7)$$

where, $\Phi(\cdot)$ and $\phi(\cdot)$ denote the standard normal distribution and density functions.

Furthermore, Schonlau (1997) observed that the classical EI acquisition function puts too much emphasis in the local search near the optima of the fitted surrogate, so he extended the classical EI to the generalized EI (GEI), of which formula is shown as below:

$$EI(\mathbf{x})^g = s^g \sum_{k=0}^g (-1)^k \binom{g}{k} u^{g-k} T_k$$

$$\begin{cases} u = (f_{\min} - \hat{y}(\mathbf{x}))/s(\mathbf{x}) \\ T_0 = \Phi(u) \\ T_1 = -\phi(u) \\ T_k = -u^{k-1}\phi(u) + (k-1)T_{k-2} \end{cases} \quad (8)$$

where the parameter g can be set as $0, 1, 2, \dots, n$; and $\binom{g}{k}$ denotes the combination. With the increase of g , the acquisition function will put more and more emphasis on the global search. In particular, the GEI is degenerated to the classical EI when $g = 1$.

Note that the GEI is proposed for single-fidelity surrogate-based optimization. However, for multi-fidelity surrogate-based optimization, a key issue would be how to adaptively distribute the new sample candidates for high- and low-fidelity simulations, respectively. To address the above sample assignment issue, we extend the GEI to Filter-GEI, which will be discussed in detail in Section 3.

3 The filter-GEI method

Different from single-fidelity surrogate-based optimization, multi-fidelity surrogate-based optimization augments the limited available high-fidelity samples by a large number of low-fidelity samples to boost the optimization performance, as shown in (9). Then, a key factor that influences the effectiveness of multi-fidelity surrogate-based optimization is

how to assign samples for high- or low-fidelity simulations in the iteration process.

$$\begin{aligned} \min \quad & y_{\text{HF}}(\mathbf{x}) \\ \mathbf{x} \in \quad & D \\ \text{with assistance of} \quad & y_{\text{LF}}(\mathbf{x}) \end{aligned} \quad (9)$$

To address the above sample assignment issue in MFO, we propose Filter-GEI, which adaptively assigns samples for HF and LF simulations in each optimization cycle. In this section, we first illustrate the details of Filter-GEI, and then we use an 1D toy problem to present the scheme of Filter-GEI intuitively.

3.1 Filter-GEI acquisition function

The MFS is often used when we can only have a small number of expensive HF samples but plenty of cheap LF samples. More importantly, when we decide to use MFS in an engineering optimization problem, we usually have some prior knowledge that the LF model can more or less capture the global trend of the HF model, even if the real optimal solutions of HF and LF models may not be exactly overlapped in the design space.

On the other hand, to take a balance in between local exploitation and global exploration in single-fidelity surrogate-based optimization, the GEI acquisition function (Schonlau 1997) often suggests a set of sample candidates either in the neighborhood of the optima of the fitted surrogate model (exploitation) or in the area of of maximum prediction uncertainty (exploration).

With the above facts in mind, we propose Filter-GEI, which calls the expensive HF simulations only when the new sample candidates have relatively better estimated objective function value, while the rest of the sample batches of GEI will be evaluated through LF simulation. In other words, we propose to impose a filter over GEI, thereby we can query HF samples to be more focused on exploiting the “promising” areas with relatively better objective function values. In the meantime, we query the other sample candidates of GEI through LF simulations to explore the optimization space globally and reduce the model uncertainty. This way we can make full use of the multi-fidelity and parallel computing resources to balance in between local and global search.

3.1.1 Formulation of the filter over GEI

To filter HF samples from the original GEI sample candidates, we propose the following threshold expression in Filter-GEI:

$$T = \omega \cdot y_{\text{HF-min}} + (1 - \omega) \cdot y_{\text{HF-mean}} \quad (10)$$

where, $y_{\text{HF-min}}$ is the current best optimal solution, $y_{\text{HF-mean}}$ is a weighted averaged function value over all training samples, and ω is an adaptive weight coefficient that weighs the importance of $y_{\text{HF-min}}$ and $y_{\text{HF-mean}}$ in determining the threshold T .

Furthermore, by taking the correlations between HF and LF models into account, we propose a dynamic weight to adjust the importance of $y_{\text{HF-min}}$ and $y_{\text{HF-mean}}$ as below:

$$\omega = \frac{1}{1 + (\hat{\sigma}_{\text{DF}}^2 / \hat{\sigma}_{\text{LF}}^2)^{0.5}} \quad (11)$$

where, σ_{DF}^2 and σ_{LF}^2 are the Gaussian process variance of DF and LF functions.

When the HF and LF models are highly correlated, the value of $\hat{\sigma}_{\text{DF}} / \hat{\sigma}_{\text{LF}}$ should be small, and the corresponding co-kriging surrogate can be expected to be in good accuracy. Then, the weight of $y_{\text{HF-min}}$ will become much higher. In other words, the Filter-GEI will make the HF simulations to be even more focused on the local exploitation of “promising” areas with relatively better estimated objective function values. On the contrary, when the correlations HF and LF models are not high, we will add more weight on $y_{\text{HF-mean}}$. In other words, the Filter-GEI will query HF samples in a much larger area to avoid the misleading of LF model and enhance the global exploration as well.

In addition, in order to take the convergence history into account, we propose a weighted averaged function value over all samples as shown in (12):

$$y_{\text{HF-mean}} = \frac{1}{\text{iter}} \sum_{i=0}^{\text{iter}} \left(\frac{1}{n_i} \sum_{j=1}^{n_i} y_{\text{HF-}ij} \right) \quad (12)$$

where, iter denotes the number of total iterations thus far, n_i is the number of newly added HF samples in the i^{th} iteration, and $y_{\text{HF-}ij}$ denotes one of the HF sample that was added in the i^{th} iteration. From (12), we can see the weights of the samples that added in the same iteration are the same, while the sample weights of different iterations vary. More importantly, the sample weights are proportional to the number of newly added samples in an iteration. Such treatment is helpful to smooth the history curve of T in (10).

With the above, the filter function we proposed in (10) can be adaptively changed, which helps to take a balance in between global and local search when assigning the HF and LF samples.

3.1.2 Sample clustering

Note that, when using the GEI acquisition function to select new sample candidates, the suggested samples of GEI with different g values may get overlapped (Li et al. 2017), resulting in ill-conditioning of the covariance matrix. To avoid the above issue and thus ensure the normal implementation of Filter-GEI-based MFO, we use hierarchical

clustering like the work in Ponweiser et al. (2008). The steps of hierarchical clustering are listed as below:

Algorithm 1 Filter-GEI based MFO algorithm.

Input: MF surrogate(contained all parameters), n

Output: X_{HF} (the HF sample set); X_{LF} (the LF sample set)

```

1: function GEI_SEARCH( $\hat{Y}_m$ )
2:    $n \leftarrow 10$ 
3:   for  $i = 1 \rightarrow n$  do
4:      $g \leftarrow i - 1$ 
5:      $[X_{c_i}, Y_{c_i}] \leftarrow \text{argmax}[E(I^g)]$ 
6:   end for
7:   return  $[X_{c_i}, Y_{c_i}] \quad i = 1 \dots n$ 
8: end function
9:
10: function GET_THRESHOLD( $\hat{Y}_m$ )
11:    $\text{ratio} \leftarrow \hat{\sigma}_d / \hat{\sigma}_{\text{LF}}$ 
12:    $\omega \leftarrow 1 / (1 + \text{ratio})$ 
13:    $y_{\min} \leftarrow \min[y_{\text{sample}}]$ 
14:    $y_{\text{mean}} \leftarrow \text{mean}[y_{\text{sample}}]$ 
15:    $T \leftarrow \omega y_{h-\min} + (1 - \omega) y_{h-\text{mean}}$ 
16:   return  $T$ 
17: end function
18:
19: function CLUSTER( $X, \text{term}$ )
20:    $[a, b] \leftarrow \text{argmin}[\text{dist}(x_i, x_j)]$ 
21:    $\text{dist}_{\min} \leftarrow \text{dist}(x_a, x_b)$ 
22:   while do  $\text{dist}_{\min} < \text{term}$ 
23:     remove point in  $x_a$  and  $x_b$ 
24:     who is closer to the existing samples
25:      $[a, b] \leftarrow \text{argmin}[\text{dist}(x_i, x_j)] \quad x_i, x_j \in X$ 
26:      $\text{dist}_{\min} \leftarrow \text{dist}(x_a, x_b)$ 
27:   end while
28:   return  $X_{\text{left}}$ 
29: end function
30:
31: function FILTER( $\hat{Y}_m$ )
32:    $[X_{c_i}, Y_{c_i}] \leftarrow \text{GEI\_search}\{\hat{Y}_m\}$ 
33:    $T \leftarrow \text{get\_threshold}\{\hat{Y}_m\}$ 
34:    $j \leftarrow 0$ 
35:    $k \leftarrow 0$ 
36:   for  $i = 1 \rightarrow n$  do
37:     if  $Y_{c_i} < T$  then
38:        $j \leftarrow j + 1$ 
39:        $[X_{h_j}, Y_{h_j}] \leftarrow [X_{c_i}, Y_{c_i}]$ 
40:     else
41:        $k \leftarrow k + 1$ 
42:        $[X_{l_k}, Y_{l_k}] \leftarrow [X_{c_i}, Y_{c_i}]$ 
43:     end if
44:   end for
45:    $X_{\text{HF}} \leftarrow \text{cluster}\{X_{\text{HF}}^{\text{initial}}, d_{\text{HF}}^T\}$ 
46:    $X_{\text{LF}}^{\text{initial}} \leftarrow \{x | x \in X_{c_i} \text{ and } x \notin X_{\text{HF}}\}$ 
47:    $X_{\text{LF}} \leftarrow \text{cluster}\{X_{\text{LF}}^{\text{initial}}, d_{\text{LF}}^T\}$ 
48:   return  $X_{\text{HF}}, X_{\text{LF}}$ 
49: end function

```

- (1) Treat each sample candidate as an independent class and calculate the minimum distance between classes;
- (2) Combine the two classes with the smallest distance into a new class;
- (3) Recalculate the distance between the new class and all classes;
- (4) Repeat (2) and (3) until the minimum distance between classes is larger than some value d^T .

Furthermore, by taking the cost of HF and LF samples into account, we use the following termination conditions when clustering samples for HF and LF, respectively:

$$\begin{aligned} d_{\text{HF}}^T &= 0.1\sqrt{d} \\ d_{\text{LF}}^T &= d_{\text{HF}}^T \left(\frac{\text{cost}_{\text{LF}}}{\text{cost}_{\text{HF}}} \right)^{\frac{1}{3}} \end{aligned} \quad (13)$$

where d denotes the dimension of the problem.

Combining the filter in (10) and sample clustering, the procedure of Filter-GEI is carried out as follows: After obtaining the sample candidates using the GEI in an optimization cycle, we first select the HF samples through (10), and we control the number of new HF samples to query through sample clustering. After that, we select the LF samples from the rest of GEI sample candidates. Based on Filter-GEI, we propose a new MFO algorithm as shown in Fig. 1. The related pseudo code is also presented in Algorithm 1.

3.2 An illustrative example

In this subsection, we use a one-dimensional toy problem to show the scheme of Filter-GEI-based MFO intuitively. Equation (14) shows the mathematical expressions of the toy problem, and we carry out optimization for this toy function as a minimization problem.

$$\begin{aligned} y_{\text{HF}} &= (6x - 2)^2 \sin(12x - 4) \\ y_{\text{LF}} &= 0.5y_{\text{HF}} + 10(x - 0.5) - 5 \\ x &\in [0, 1] \end{aligned} \quad (14)$$

Initially, we use 3 HF samples and 5 LF samples to build a co-kriging surrogate, and then we use the Filter-GEI algorithm to carry out the optimization, as shown in Fig. 2. Table 1 lists the sample details in each optimization cycle.

In Fig. 2, the hollow squares denote the original sample candidates obtained from GEI, while the hollow circles denote the new samples to be queried through LF simulation, the threshold to filter HF samples is represented by dashed straight lines. As shown in Fig. 2a, the sample candidates originally suggested by GEI are mainly gathered around $x = 0.26$ and $x = 0.70$. Through the filter using (10) and sample clustering as well, the sample $x=0.7053$ is selected for HF simulation in this iteration. Thereafter, we cluster the rest GEI sample candidates and select $x = 0.6504$

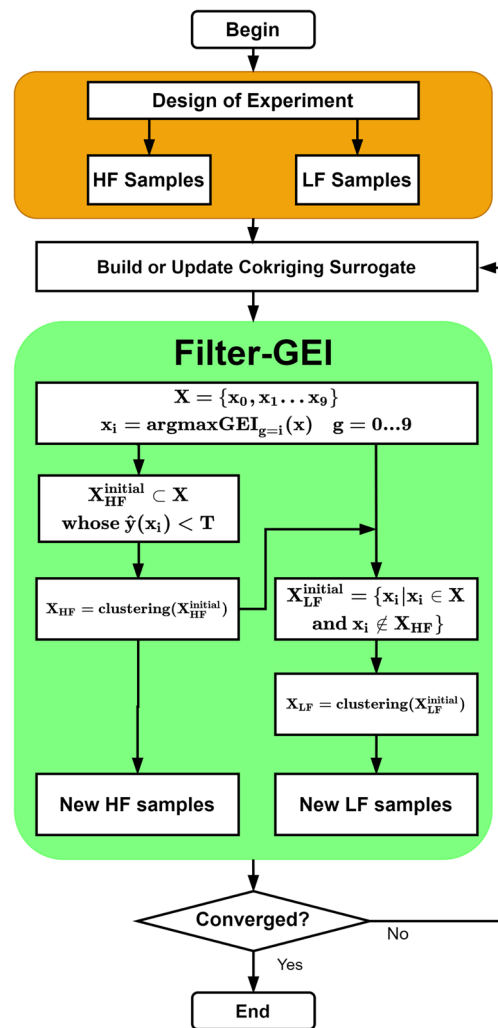


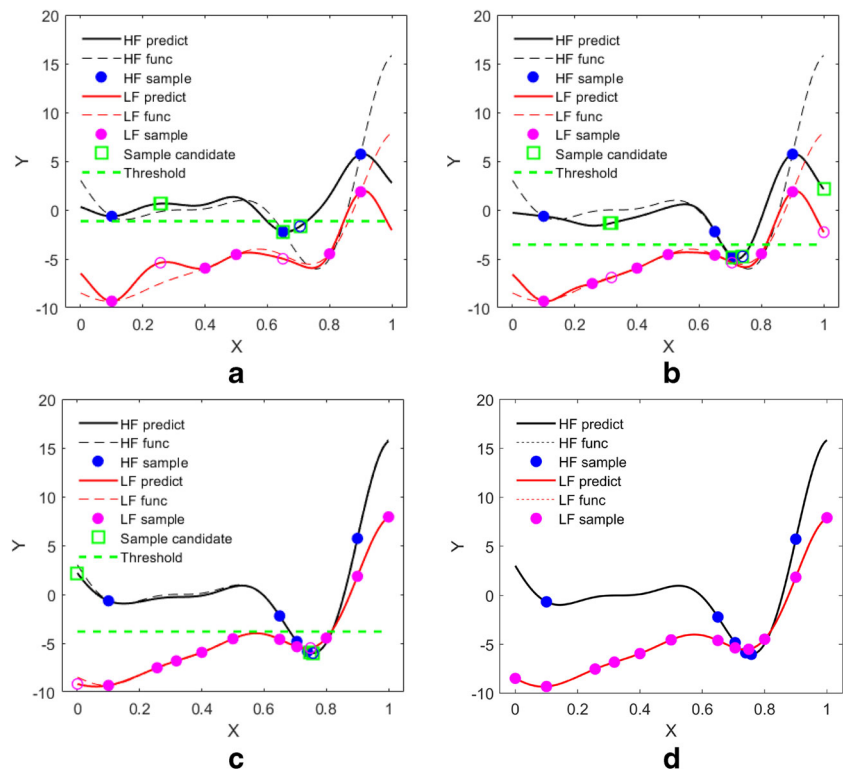
Fig. 1 Flowchart of the Filter-GEI Algorithm

and $x=0.2562$ to be evaluated through LF simulation in this optimization cycle.

Similarly, in the second iteration (see Fig. 2b), the Filter-GEI assigns the boundary sample candidate as LF samples to reduce the uncertainty of co-kriging, while the HF simulation is called to exploit the “promising” area with relatively better estimated objective function value. Finally, as shown in Fig. 2d, the Filter-GEI successfully finds the real optimal solution of the toy problem. In the meantime, the HF function is perfectly fitted by the co-kriging, even though no samples are located at the boundaries of the HF curve.

With the above, the motivation of Filter-GEI is well demonstrated. That is, while balancing in between the local exploitation and global exploration, we can filter HF samples to be more focused on exploiting “promising” areas with relatively better objective function value. In the meantime, considering the correlations between HF and LF models, we can query more LF samples in the rest of

Fig. 2 The optimization process of the one-dimensional toy function by using Filter-GEI. **(a)** iteration 1. **(b)** iteration 2. **(c)** iteration 3. **(d)** well fitted



the areas to reduce the model uncertainty over space. This way we can make full use of multi-fidelity and parallel computing resources to achieve the optimal solution most efficiently and effectively.

4 Experimental tests

In this section, we test the proposed Filter-GEI algorithm and compare it with a set of the state of the art MFO algorithms on both benchmark functions and an engineering optimization design problem.

4.1 Baseline for comparison and experimental setup

To verify the effectiveness of Filter-GEI, we compare it with two state of the art algorithms as augmented-EI (Huang et al. 2006) and VF-EI (Zhang et al. 2018). Specifically,

Table 1 The sampling detail of illustrative optimization example

Iteration	HF points	LF points	T
Initial	[0.1;0.65;0.9]	[0.1;0.4;0.5;0.8;0.9]	\
1	0.7053	[0.2562;0.6504]	-1.160
2	0.7395	[0.3180;0.7053;1.0000]	-3.529
3	0.7574	[0.0000;0.7490]	-3.808

the augmented-EI is the first MFO algorithm which queries both HF and LF samples in the iteration process. It takes both the similarities and the cost ratio between HF and LF samples into account when deciding whether to call the HF or LF simulation. On the other hand, VF-EI is a recently proposed MFO algorithm, which takes a special consideration on the model uncertainty to improve the performance of MFO.

Note that both augmented-EI and VF-EI add one sample in each optimization cycle. Differently, our proposed Filter-GEI can be first parallel MFO algorithm, which adaptively assign sample batches for HF and LF simulation in each iteration. The comparisons between Filter-GEI and augmented-EI and VF-EI shall manifest the advantage of parallel optimization in accelerating the process.

To show the role of filtering of HF samples, we also compare Filter-GEI with the GEI-based MFO, which directly add the sample batches of GEI for both HF and LF simulations in each iteration. In addition, we also test EI-based MFO, which add in one sample for both HF and LF evaluations per optimization cycle. Note that, infilling both LF and HF samples at the same location is helpful to improve the accuracy of discrepancy function, though it cannot provide more information of HF model in the same location.

For fair comparison, we use the open-source code oDACE (Couckuyt et al. 2014) to build a uniform test bed for all the comparing algorithms. For each testing case, the

Table 2 Definition of multi-fidelity test cases

No.	HF function	LF function	Dimension	Termination error
Case1	Forrester1	Forrester1a	1	0.0602 (1%)
Case2	Forrester1	Forrester1b	1	0.0602 (1%)
Case3	Hartman3	Hartman3+MA3*7.6	3	0.0386 (1%)
Case4	Ackley5	Ackley5+MA5*0.74	5	0.1000 (optimal value is 0)
Case5	Hartmann6	Hartmann6+MA6*0.21	6	0.0304 (1%)

number of initial HF and LF samples is set as three and six times the function dimension, respectively. We use the strategy of nearest neighbor sampling (Guo et al. 2018) to generate these HF and LF samples.

4.2 Tests and discussions on benchmark functions

We use the benchmark functions in Zhang et al. (2018) to carry out the optimization tests; the definitions and

Table 3 Description of test function

Name	Description
Forrester1	$f(x) = (6x - 2)^2 \sin(12x - 4)$ $x \in [0, 1], \quad x^* = 0.7525, \quad f^* = -6.020740$
Forrester1a	$f(x) = 0.5f(x) + 10(x - 0.5) - 5$
Forrester1b	$f(x) = f(x) - 5$
Hartman3 (Hartman 1973)	$f(x) = -\sum_{i=1}^4 c_i \exp[-\sum_{j=1}^3 \alpha_{ij}(x_j - p_{ij})^2]$ where $\alpha = \begin{bmatrix} 3 & 10 & 30 \\ 0.1 & 10 & 35 \\ 3 & 10 & 30 \\ 0.1 & 10 & 35 \end{bmatrix}, c = \begin{bmatrix} 1 \\ 1.2 \\ 3 \\ 3.2 \end{bmatrix}, p = \begin{bmatrix} 0.3689 & 0.1170 & 0.2673 \\ 0.4699 & 0.4387 & 0.7470 \\ 0.1091 & 0.8732 & 0.5547 \\ 0.0381 & 0.5743 & 0.8828 \end{bmatrix}$ $x \in [0, 1], \quad x^* = (0.114, 0.556, 0.852), \quad f^* = -3.8627$
MA3 (McDaniel 2000)	$f(x) = 0.585 - 0.324x_1 - 0.379x_2 - 0.431x_3 - 0.208x_1x_2 + 0.326x_1x_3$ $+ 0.193x_2x_3 + 0.225x_1^2 + 0.263x_2^2 + 0.274x_3^2$
Ackley5	$f(x) = -a \exp\left[-b\sqrt{\frac{1}{n}\sum_{i=1}^5 x_i^2}\right] - \exp\left[\frac{1}{n}\sum_{i=1}^5 \cos(cx_i)\right] + a + \exp(1)$ $a = 20; b = 0.2; c = 2\pi$ $x_i \in [-2, 2], \quad x^* = (0, 0, 0, 0, 0), \quad f^* = 0$
MA5	$f(x) = 0.585 - 0.00127x_1 + 0.00113x_2 + 0.00663x_3 + 0.0129x_4 + 0.00611x_5$ $+ 0.00526x_1x_4 + 0.0106x_1x_5 - 0.000626x_2x_4 - 0.00310x_2x_5$ $- 0.00724x_4x_5 + 0.00096x_3^2 + 0.0124x_4^2 + 0.0101x_5^2$
Hartmann6	$f(x) = -\sum_{i=1}^4 c_i \exp[-\sum_{j=1}^3 \alpha_{ij}(x_j - p_{ij})^2]$ where $\alpha = \begin{bmatrix} 10 & 3 & 17 & 3.50 & 1.7 & 8 \\ 0.05 & 10 & 17 & 0.1 & 8 & 14 \\ 3 & 3.5 & 1.7 & 10 & 17 & 8 \\ 17 & 8 & 0.05 & 10 & 0.1 & 14 \end{bmatrix}, c = \begin{bmatrix} 1 \\ 1.2 \\ 3 \\ 3.2 \end{bmatrix}$ $p = 10^{-4} \begin{bmatrix} 1312 & 1696 & 5569 & 124 & 8283 & 5886 \\ 2329 & 4135 & 8307 & 3736 & 1004 & 9991 \\ 2348 & 1451 & 3522 & 2883 & 3047 & 6650 \\ 4047 & 8828 & 8732 & 5743 & 1091 & 381 \end{bmatrix}$ $x \in [0, 1], \quad x^* = (0.20169, 0.150011, 0.476874, 0.275332, 0.311652, 0.6573), \quad f^* = -3.04246$
MA6	$f(x) = 0.544 + 0.15984x_1 + 0.18857x_2 + 0.02371x_3 - 0.04144x_4 + 0.3313x_5$ $- 0.11813x_6 - 0.03719x_1x_2 + 0.03099x_2x_3 - 0.0504x_3x_6 - 0.03103x_4x_5$ $+ 0.19345x_1x_6 - 0.2635x_1^2 - 0.17694x_2^2 - 0.23528x_5^2$

descriptions of the benchmark functions are shown in Tables 2 and 3. As the real optimal solution of benchmark functions are known, the comparing algorithms shall be terminated when the relative error between the obtained optimal solution and the real one is less than 1% or the number of iterations is 15 times the function dimension. As an exception, since the optimal solution of case 4 in Table 2 is 0, we follow the treatment in Zhang et al. (2018), i.e., the algorithms will be stopped once the absolute error between the obtained optimal solution and the real one was less than 0.1. In addition, each benchmark function will run 20 times; the mean and standard deviation of the obtained optimal solutions will be calculated to compare the performance of different algorithms.

Table 4 shows the comparison of the results of the five benchmark functions. When comparing between Filter-GEI with GEI- and EI-based MFO, the Filter-GEI always uses the minimum number of HF samples to achieve the competitive optimal solution. As a contrast, without filtering of HF samples, the GEI-based MFO needs significantly more HF samples. In the meantime, as the Filter-GEI- and GEI-based MFO add multiple samples instead of one

sample in each iteration, these two algorithms usually need much less iterations than EI-based MFO to meet the termination condition. Meanwhile, the Filter-GEI consumes the same or even less HF samples than EI-based MFO to achieve the competitive results. To summarize, the Filter-GEI outperforms GEI-based MFO in terms of the number of HF samples, and it has better efficiency than EI-based MFO in terms of both iterations and the number of HF samples.

When comparing the VF-EI and augmented-EI, Filter-GEI needs significant fewer iterations to achieve comparable optimal solution. More importantly, as shown in the last column of Table 4, the VF-EI only succeeds to achieve the optimal solution 12 times out of 20 tests for the five-dimensional Ackley5 function (case 4), and the augmented-EI has a success rate of 12/20 for the six-dimensional Hartman6 function (case 5). On the contrary, Filter-GEI succeeds to find the optimal solution for all the testing cases. In the meantime, as shown in the first column of Table 4, the fluctuations of iterations of Filter-GEI are much smaller than that of VF-EI and augmented-EI. The above results indicate that, in addition to having significantly better efficiency in terms of wall-clock time (i.e., the number of

Table 4 Optimization results of Filter-GEI, augmented-EI, EI, GEI, and VF-EI on five benchmark functions

Function	Algorithms	Iteration	Optimal	HF	LF	Success rate
case1	Filter-GEI	2 ± 1	-6.0103 ± 0.0146	6 ± 1	11 ± 2	20/20
	GEI	3 ± 1	-6.0111 ± 0.0171	10 ± 1	11 ± 3	20/20
	EI	3 ± 1	-6.0104 ± 0.0165	6 ± 1	9 ± 1	20/20
	VF-EI	5 ± 2	-6.0150 ± 0.0150	6 ± 2	8 ± 1	20/20
	augmented-EI	5 ± 3	-6.0117 ± 0.0115	6 ± 2	8 ± 1	20/20
case2	Filter-GEI	3 ± 1	-6.0115 ± 0.0142	6 ± 1	11 ± 2	20/20
	GEI	3 ± 1	-6.0044 ± 0.0199	11 ± 3	11 ± 4	20/20
	EI	3 ± 1	-6.0082 ± 0.0136	6 ± 1	9 ± 1	20/20
	VF-EI	5 ± 2	-6.0191 ± 0.0058	5 ± 2	9 ± 1	20/20
	augmented-EI	3 ± 1	-6.0114 ± 0.0131	5 ± 1	7 ± 1	20/20
case3	Filter-GEI	4 ± 1	-3.8450 ± 0.0107	16 ± 2	39 ± 5	20/20
	GEI	4 ± 1	-3.8491 ± 0.0093	27 ± 4	32 ± 9	20/20
	EI	9 ± 4	-3.8457 ± 0.0101	18 ± 4	27 ± 4	20/20
	VF-EI	13 ± 5	-3.8507 ± 0.0099	16 ± 4	24 ± 4	20/20
	augmented-EI	10 ± 3	-3.8511 ± 0.0103	15 ± 2	21 ± 2	20/20
case4	Filter-GEI	20 ± 3	0.0770 ± 0.0164	37 ± 3	92 ± 13	20/20
	GEI	16 ± 4	0.0744 ± 0.0178	64 ± 4	76 ± 12	20/20
	EI	25 ± 6	0.0713 ± 0.0183	40 ± 6	55 ± 6	20/20
	VF-EI	47 ± 24	0.4945 ± 0.9972	16 ± 0	76 ± 24	12/20
	augmented-EI	30 ± 14	0.0673 ± 0.0223	28 ± 5	47 ± 11	20/20
case5	Filter-GEI	12 ± 4	-3.0312 ± 0.0081	33 ± 6	94 ± 24	20/20
	GEI	17 ± 9	-3.0115 ± 0.0215	123 ± 91	141 ± 101	14/20
	EI	25 ± 22	-3.0176 ± 0.0147	43 ± 22	61 ± 22	18/20
	VF-EI	42 ± 30	-3.0216 ± 0.0263	23 ± 4	73 ± 29	15/20
	augmented-EI	51 ± 33	-3.0058 ± 0.0231	36 ± 8	69 ± 30	12/20

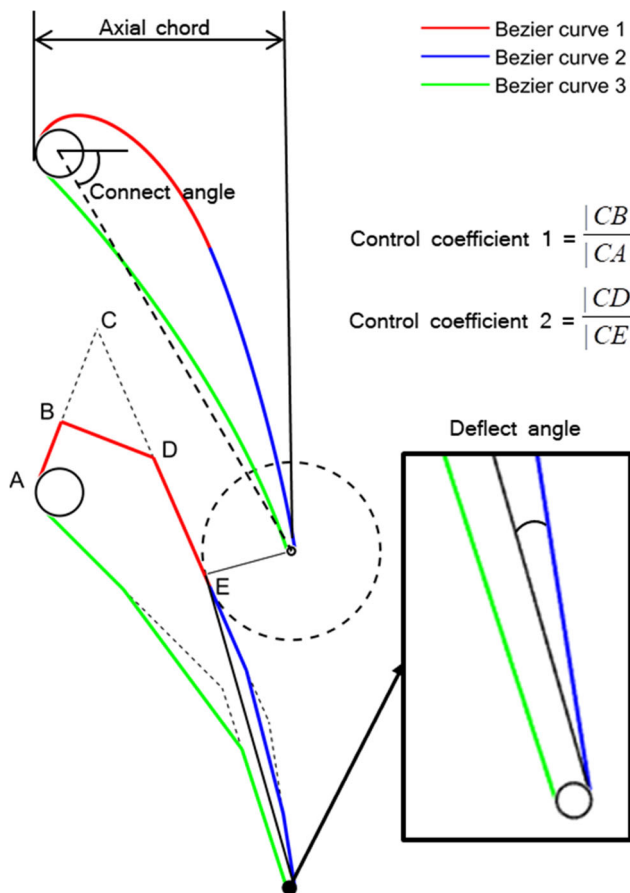


Fig. 3 Parameterization method for the engineering design

iterations rather than the number of simulations), the performance of Filter-GEI is also much more robust than those of VF-EI and augmented-EI. With the above, the effectiveness of Filter-GEI and its advantage over other the state of the art algorithms have been well demonstrated.

4.3 Engineering case study

In this subsection, we test Filter-GEI on a turbine design problem to further validate its effectiveness.

Table 5 Design variables in the engineering design case

No.	Geometric definition	Reference value	Variation range
1	axial chord	33.9[°]	(−2.0, 2.0)
2	center connect angle	59.8[°]	(−4.0, 2.0)
3	inlet upper wedge angle	69.0[°]	(−15.0, 3.0)
4	outlet deflect angle	4.5[°]	(−1.5, 4.5)
5	correlation coefficient	0.35[-]	(−0.05, 0.10)
6	control coefficient 1	0.40[-]	(−0.15, 0.15)
7	control coefficient 2	0.50[-]	(−0.15, 0.15)

Table 6 Design conditions

Condition name	Value
inlet total temperature[°C]	709
inlet total pressure[Pa]	344740
inlet flow angle[°]	90
Mach number at cascade outlet [-]	0.878

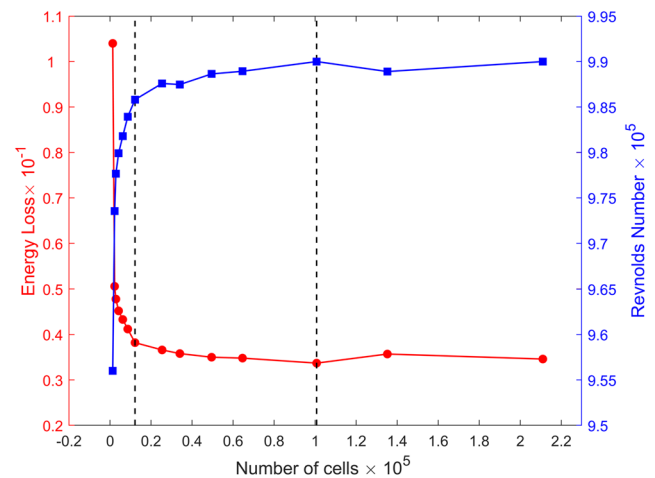


Fig. 4 Tests on grid independence for the engineering problem

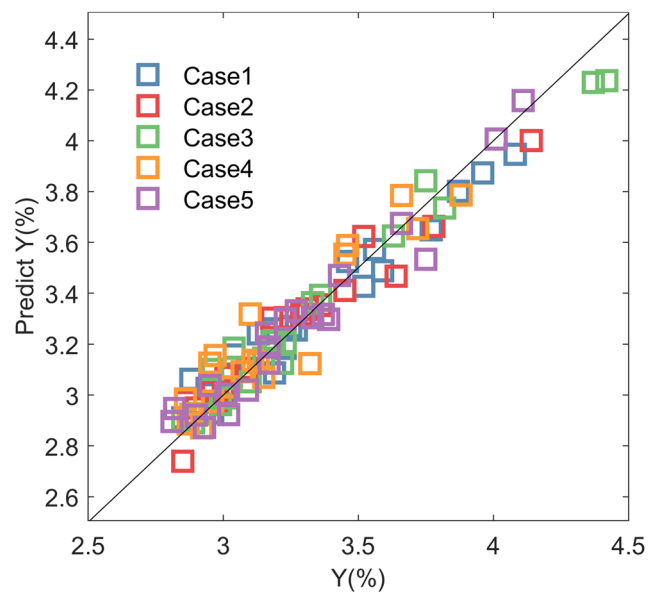


Fig. 5 Scatterplot of the cross validation results for the engineering problem

Table 7 Cross-validated results of co-kriging model in the five tests for the engineering problem

Surrogate		Case1	Case2	Case3	Case4	Case5
co-kriging	CV-RMSE	9.21E-02	1.12E-01	7.87E-02	3.03E-01	7.87E-02
	CV-R ²	0.938	0.952	0.970	0.862	0.953

4.3.1 Problem description

We select the first-stage stator vane of the well-known Energy Efficient Engine (GE-E³) (Cherry and Dengler 1984) turbo for the test. The goal is to minimize the energy loss of the stator vane through optimizing the contour of stator vane.

We use the Bezier curves to parameterize the stator vane as shown in Fig. 3. The profile is composed of three Bezier curves and two arcs. Each Bezier curve is adjusted by four control points, and the variations of the 12 control points are determined by 18 design parameters. In this study, we select seven variables from the 18 design parameters to optimize the stator profile. The definition and variable range of these seven variables are shown in Table 5.

4.3.2 High- and low-fidelity simulation models

The definition of the energy loss is shown in (15):

$$\xi = \left(1 - \frac{h_{1-1} + 0.5c_{1-1}^2}{h_{0-0} + 0.5c_{0-0}^2} \right) \times 100\% \quad (15)$$

where h denotes the enthalpy, and c denotes the flow velocity at different locations of stator vane. The subscripts 0–0 and 1–1 denote the inlet and outlet of the stator, respectively.

To obtain the energy loss of stator vane, we use the commercial software ANSYS CFX 14.5 to carry out the HF and LF simulations. Specifically, we employ the Reynolds Averaged Navier-Stokes Equations (RANS) coupled by the Spalart-Allmaras turbulence model for the simulation process. The boundary conditions at the inlet and outlet of the stator are shown in Table 6, by referring to the descriptions of GE-E³ turbine in Cherry and Dengler (1984). The grid of stator is automatically generated using one O-grid block and three H-grid blocks by using an in-house code, and Fig. 4 presents the test on the grid independence.

Table 8 Optimization results after adding 35 HF samples

	Iteration	Optimal(%)	Min(%)	Max(%)	HF number	LF number
Filter-GEI	13 ± 1	2.720 ± 0.0245	2.70	2.76	56	91 ± 10
GEI	8 ± 0	2.760 ± 0.0224	2.73	2.79	56	79 ± 2
EI	36 ± 0	2.732 ± 0.0192	2.70	2.75	56	77 ± 0
VF-EI	45 ± 3	2.748 ± 0.0239	2.72	2.78	56	51 ± 3
augmented-EI	51 ± 6	2.726 ± 0.0358	2.69	2.78	56	59 ± 9

Similar to Zhang et al. (2018), the HF and LF simulation models are obtained by using different cell numbers. In this study, the cell numbers in HF and LF models are set as 100,701 and 12,141, respectively. Correspondingly, the cost ratio of LF and HF models is 0.1.

4.3.3 Results and discussions

For the turbine design optimization with seven design variables, we initially generate 42 LF samples and 21 HF samples by using the nearest neighbor sampling (Guo et al. 2018). Different from the benchmark functions, the optimal solution of a real-world problem is often unknown; therefore, we compare the performance of the different algorithms by fixing the total number of HF samples, i.e., the comparing algorithms will be terminated after consuming 35 HF samples in the iteration process. Considering the expensive cost of simulations, we run the engineering case five times to compare the performance of different algorithms.

Before carrying out the optimization process, we use the leave-one-out validation (Loo-CV) to validate the accuracy of co-kriging, as shown in Fig. 5. Table 7 shows the quantitative measures for the five tests. The definitions of cross-validated root mean squared error (CV-RMSE) and cross validated R^2 (CV- R^2) are shown in (16).

$$\begin{aligned} \text{CV-RMSE} &= \sqrt{\sum_{i=1}^N (y_i - \hat{y}_i)^2 / N} \\ \text{CV-}R^2 &= 1 - \sum_{i=1}^N (y_i - \hat{y}_i)^2 / \sum_{i=1}^N (y_i - \bar{y})^2 \end{aligned} \quad (16)$$

In Fig. 5, the surrogate prediction values $\hat{y}$ and the true function y are distributed closely to the line $y = \hat{y}$, which indicates that the co-kriging models we built were in good accuracy. In Table 7, all the cross-validated R^2 are larger than 0.85, and most of them are larger than 0.95, which further confirms the good accuracy of co-kriging models.

Table 8 shows the optimization results after consuming 35 HF samples, and Fig. 6 shows the averaged convergence

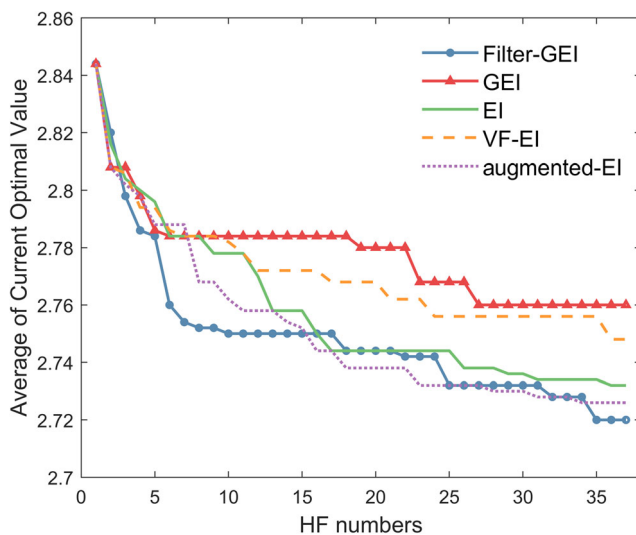


Fig. 6 Convergence history from the view of HF numbers

history of different algorithms. Similar to the results of benchmark functions, Filter-GEI and GEI-based MFO use significant fewer iterations to obtain the optimal solution. More importantly, as shown in Fig. 6, with the filter of HF samples, the Filter-GEI shows significantly faster convergence rate than the GEI-based MFO. The Filter-GEI also achieves the best optimal solution after consuming 35 HF samples. With the above, the effectiveness of Filter-GEI is further demonstrated, which can make full use of multi-fidelity and parallel computing resources to achieve the optimal solution most efficiently.

5 Conclusion

In this paper, we propose a new infill criterion named as Filter-GEI for multi-fidelity surrogate-based optimization (MFO). Different from previous works, we propose to infill multiple high-fidelity (HF) and low-fidelity (LF) samples rather than one sample per optimization cycle, in order to make use of the multi-fidelity and parallel computing resources. More specifically, considering the correlations between HF and LF simulations, we propose an adaptive filter function on top of the generalized expected improvement (GEI) acquisition function, and we name it Filter-GEI. The Filter-GEI can adaptively filter the HF samples to be evaluated. Thereby, we can call more HF samples to exploit “promising” areas with relatively better objective function value, while we query LF samples to explore the rest of areas globally. This way we achieve a good balance in between the local and global search by taking full advantage of multi-fidelity and parallel computing resources. Through tests on five benchmark functions and one engineering problem of the

turbine blade design, the proposed Filter-GEI outperforms the compared algorithms. Hence, the effectiveness of our proposed algorithm is well demonstrated.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Replication of results The replication of results can be found through the web link: <https://github.com/zhet1997/Filter-GEI.code>

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